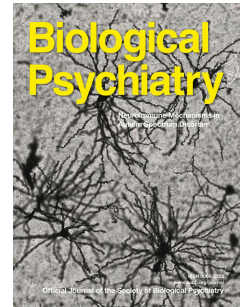


Journal Pre-proof

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PII: S0006-3223(24)01556-7

DOI: <https://doi.org/10.1016/j.biopsych.2024.08.023>

Reference: BPS 15584

To appear in: *Biological Psychiatry*

Received Date: 29 February 2024

Revised Date: 20 August 2024

Accepted Date: 21 August 2024

Please cite this article as: Riegner G., Dean J., Wager T.D. & Zeidan F., Mindfulness meditation and placebo modulate distinct multivariate neural signatures to reduce pain, *Biological Psychiatry* (2024), doi: <https://doi.org/10.1016/j.biopsych.2024.08.023>.

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TITLE

Mindfulness meditation and placebo modulate distinct multivariate neural signatures to reduce pain

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Acknowledgements: This work was funded by the National Center for Complementary and Integrative Health.

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Keywords: pain, mindfulness, meditation, placebo, fMRI, MVPA

Short title: Mindfulness and placebo engage distinct brain systems

ABSTRACT

Background: Rather than a passive reflection of nociception, pain is shaped by the interplay between one's experiences, current cognitive-affective states, and expectations. The placebo-response, a paradoxical yet reliable phenomenon, is postulated to reduce pain by engaging mechanisms shared with "active" therapies. It has been assumed that mindfulness meditation, practiced by sustaining nonjudgmental awareness of arising sensory events, merely reflects mechanisms evoked by placebo. Recently, brain-based multivariate pattern analysis (MVPA) has been validated to successfully disentangle nociceptive-specific, negative-affective, and placebo-based dimensions of the subjective pain experience. **Methods:** To determine if mindfulness meditation engages distinct brain mechanisms from placebo and sham-mindfulness to reduce pain, MVPA pain signatures were applied across two randomized clinical trials that employed overlapping psychophysical pain testing procedures (49°C noxious heat; visual analogue pain scales) and distinct fMRI techniques (blood-oxygen-level dependent; perfusion-based). After baseline pain testing, 115 healthy participants were randomized into a four-session mindfulness meditation (n = 37), placebo-cream conditioning (n = 19), sham-mindfulness meditation (n = 20), or book-listening (n = 39) intervention. After each intervention, noxious heat was administered during fMRI and each manipulation. **Results:** A double dissociation in the MVPA signatures supporting pain regulation was revealed by mindfulness meditation as compared to placebo-cream. Mindfulness meditation produced significantly greater reductions in pain intensity and pain unpleasantness ratings, nociceptive-specific and negative-affective pain signatures when compared to placebo-cream, sham-mindfulness meditation and controls. Placebo-cream only reduced the placebo-based signature. **Conclusions:** Mindfulness meditation and placebo engage distinct neural pain signatures to reduce pain to demonstrate mechanistic granularity between placebo and mindfulness.

INTRODUCTION

The placebo response is characterized as an “inactive” manipulation that reliably promotes analgesia. The nature of the placebo effect is paradoxical and, to date, has been shown to engage mechanisms that modulate beliefs, expectations, and conditioning to attenuate the subjective experience of pain. To this extent, placebo is the gold standard comparator in evaluating the efficacy and mechanisms of pain-related therapies (1-3). There is mounting evidence from several randomized clinical trials demonstrating that mindfulness meditation produces lasting improvements in chronic pain (4-6). However, appropriate placebo-controlled comparators are rarely employed in pain-focused mindfulness studies.

In brief, mindfulness meditation is a self-regulated cognitive practice that is predicated on developing a non-evaluative and present-centered approach to a subjectively available pain experience. As adapted in our laboratory, mindfulness meditation is practiced by focusing on the breath sensations in a non-evaluative fashion. Converging lines of contemplative text (7) and scientific evidence (8-11) indicate that mindfulness-induced pain relief is driven by a) the attenuation of neural activity in lower-level supraspinal regions (contralateral thalamus and somatosensory cortices) (11-13), b) deactivation of cortical midline structures [precuneus and ventromedial prefrontal cortex (vmPFC)] supporting self-referential processing and affective meaning (11-15), and c) non-opioidergically mediated processes (16-18). In contrast, placebo-analgesia is largely mediated by endogenous opioids (19) and greater dorsolateral prefrontal cortical (dlPFC) and periaqueductal gray matter (PAG) activation (20).

Recent findings show that mindfulness meditation engages both distinct and overlapping endogenous (16, 17) and neural (13) mechanisms compared to placebo. While mindfulness and placebo might modulate similar brain systems to reduce pain, conventional univariate neuroimaging analyses used in these studies a) lack specificity due to the reverse inference problem (18), b) cannot reliably parcel out stimulus driven nociceptive processing from pain-related appraisals (21), and/or c) may be unable to explicitly isolate neural processes supporting meditation from placebo. Instead, multivariate models can identify the specific brain patterns linked to nociceptive and extra-nociceptive (evaluative) dimensions of pain (22). This application of supervised machine learning, termed multivariate/multivoxel pattern analysis (MVPA), can capture the distributed nature of pain-related neural processing and predict subjective pain reports with high accuracy. Thus, MVPA pain signatures were used to isolate and identify the specific brain systems supporting mindfulness meditation as compared to placebo-induced analgesia.

Functional magnetic resonance imaging (fMRI)-based MVPA has been successful at disambiguating pain-evoked brain activity linked to nociception (23), manipulations of expectations (placebo) (24), and negative affect (25). In brief, the Neurologic Pain Signature (NPS) reliably tracks the intensity of pain elicited by increases in noxious stimulus intensity (23), but is largely insensitive to placebo (26) and several other psychological influences on pain (24). For these pain signatures, positive and negative voxel weights correspond to increased and decreased pain, respectively. The NPS pattern weights are positive in the thalamus, anterior-mid-posterior insula, cerebellum, PAG and dorsal anterior cingulate cortex (dACC), and negative in the vmPFC, precuneus and posterior cingulate cortex (PCC) (23). The Stimulus Intensity Independent Pain Signature-1 (SIIPS-1) predicts pain driven by placebo-related factors such as expectations for relief, conditioning, perceived control, and psychosocial contexts (24, 27). SIIPS-1 has negative predictive weights in vmPFC, nucleus accumbens, precuneus, hippocampus and positive predictive weights in nociceptive areas including the insula, mid-cingulate cortex, posterior insula and cerebellum (24). The Negative Affective Pain Signature (NAPS) was recently validated to reliably predict negative affect dimensions of pain (25) and has positive weights in the thalamus, somatosensory cortices (SI/SII), posterior insula, and midline and lateral PFC cortices. These signatures, however, have yet to identify if mindfulness meditation or sham-mindfulness

meditation acutely modulate pain response patterns in the brain that are distinct from those engaged by placebo.

To address this gap in knowledge, the present study combined datasets ($n = 115$ total healthy volunteers) from two previously published studies using distinct fMRI techniques—blood-oxygen-level dependent imaging (BOLD; Study 1) (11) and perfusion-based pseudo-continuous arterial spin labeling (PCASL; Study 2) (13). Given the effects of the nonspecific components of practicing mindfulness meditation (posture, breathing, beliefs), we also included a sham-mindfulness meditation condition/group, wherein meditation naïve participants were led to believe they were practicing mindfulness meditation while performing controlled slow and deep breathing. The present study examined if and how mindfulness meditation, compared to placebo-cream, sham-mindfulness meditation, and book-listening controls, modulated pain and MVPA pain signatures.

We postulated that mindfulness meditation, as compared to rest, the book-listening control, sham-mindfulness meditation and placebo, would reduce peripherally mediated nociception (NPS) and higher-order negative affect (NAPS) brain responses. Further, we have shown that mindfulness meditation related pain relief is not driven by expectations, beliefs, or conditioning and engages different brain processes than placebo (13). Thus, we did not expect that mindfulness meditation would attenuate the placebo-based SIIPS-1. Conversely, sham-mindfulness meditation reliably reduces pain and has been associated with mechanisms supporting placebo (13). Thus, we expected that the sham-mindfulness meditation would reduce the NPS and SIIPS-1. Finally, we postulated that the placebo-cream condition would reduce the SIIPS-1 but not the NPS or NAPS.

MATERIALS AND METHODS

Participants

Study 1 included 40 participants (mean age = 31 ± 10 years; 20 females) and Study 2 included 75 participants (mean age = 27 ± 6 years; 38 females). The two studies used the same recruitment call to “test the effects of cognitive techniques on pain and the brain” (see Supplementary Methods for more details).

Thermal Stimuli and Psychophysical Assessment of Pain

Study 1 and 2 used the same 16mm² Medoc TSA-II probe to deliver thermal stimuli to the back of the right calf. In Study 1, heat series (300 seconds) consisted of ten, 10-second plateaus of 49°C interleaved with 14 seconds of 35°C stimulation to allow time for hemodynamic recovery of the BOLD signal. In Study 2, the heat series (264 seconds) included a total of ten, 12-second plateaus of 49°C interleaved between 8-second periods of 35°C stimulation to increase signal to noise ratio and detect significant regional cerebral blood flow (CBF) changes. Following each series, ratings for pain intensity and pain unpleasantness were collected using an 11-point visual analog scale (VAS; 0 = “no pain” to 10 = “worst pain imaginable”). The probe was moved to a new stimulation site on the right calf to avoid habituation or sensitization after each heat series and pain rating collection.

Acquisition of MRI Data

In both studies, MRI data were acquired on the same 3T Siemens Skyra scanner with a 32-channel head coil (see Supplemental Methods for anatomical and fMRI acquisition parameters).

Experimental Design

Pre-intervention pain testing

All participants were first familiarized with the experimental stimuli and trained to use the VAS by providing pain ratings in response to thirty-two, 5-second duration stimuli (within the range of 35°C - 49°C) delivered to the left ventral forearm. Pre-intervention noxious heat series were then administered, and VAS pain ratings were collected after each respective heat series. In Study 1, four noxious heat series (49°C) were delivered to the right calf outside the scanner. Only the first two heat series are included in this study to better match Study 2's procedures. In Study 2, two noxious heat (49°C) and two innocuous (35°C) heat series were administered during PCASL fMRI acquisition (pre-intervention fMRI data previously reported (13) and are not presented here).

Randomization

After completion of the baseline pain testing session, participants were randomized into sex-stratified intervention groups by study personnel not involved in data collection. In Study 1, participants were randomized to a four-session mindfulness meditation training ($n = 20$) or book-listening control group ($n = 20$), respectively. In Study 2, participants were randomized into a four-session mindfulness meditation ($n = 17$), sham-mindfulness meditation ($n = 20$), placebo-cream conditioning ($n = 19$), and book-listening control ($n = 19$) group. In total, 37 participants were randomized to the mindfulness meditation, 39 to the book-listening control, 20 to the sham-mindfulness meditation, and 19 to the placebo-cream group.

Intervention Sessions: Study 1 and Study 2 included the same mindfulness meditation intervention and book-listening control regimen.

Mindfulness meditation group: Participants participated in four, separate 20-minute sessions of mindfulness meditation training facilitated by experienced instructors. Across all training sessions, participants were taught to focus on the changing cadence of the breath and to reduce self-referential judgments by acknowledging

arising thoughts, feelings, and emotions without reaction. When distracted, participants were told to “return their attention back to the breath.” The amount of instruction from session to session decreased sequentially, and in the final training session, an audio recording of the MRI scanner was played to simulate meditating in an MRI environment. Importantly, participants were told not to practice outside of study training and were not explicitly instructed to change their breathing rate.

Book-listening control group: The book-listening control group listened to an audio recording of *The Natural History and Antiquities of Selborne* (28) across four separate 20-minute sessions. All other experimental conditions, including setting and intervention facilitators, paralleled the meditation trainings. Participants were not allowed to talk to each other, use their cellphones and/or computers.

Placebo-cream control group (Study 2 only): In placebo conditioning, twelve, noxious heat thermal stimuli (ten, 12-second trials of 49°C) were delivered to the back of the calf. Next, a technician wearing a white laboratory coat and blue medical gloves administered the placebo cream (petrolatum jelly) to the skin. Participants were instructed that the cream is called lidocaine and “has been proven effective at progressively reducing pain after multiple applications.” Ten minutes after application, the cream was removed, and the thermal stimulus was again administered in the region of where the cream was applied. However, across the four placebo-conditioning sessions, the stimulus temperature was covertly and sequentially lowered (session 1=48°C, sessions 2-3=47°C, and session 4= 46.5°C) to enhance placebo conditioning by leading participants to believe that the cream was analgesic. VAS pain intensity and unpleasantness ratings were collected before and after each conditioning session and presented elsewhere (13).

Sham-mindfulness meditation group (Study 2 only): The main purpose of the sham-mindfulness meditation intervention was to lead subjects to believe they were practicing *mindfulness meditation without the instructions related to mindfully attending to the breath in a non-evaluative manner*. This regimen was designed so that the only difference between the mindfulness meditation and sham-mindfulness meditation group's training was the explicit mindfulness-specific instructions to practice mindful attention to the breath with non-evaluative appraisal of discursive sensory events. Participants in the sham-mindfulness meditation group were told that they were randomly assigned to the mindfulness meditation group. In each of the four training sessions (20 minutes each), participants were instructed to close their eyes, and to take a deep breath “as we sit here in meditation” every 2–3 minutes. All other aspects of the sham-mindfulness meditation intervention (posture, intervention room, facilitator, time spent providing instructions) matched the mindfulness meditation regimen. The amount of instruction from session to session decreased sequentially, and in the final training session, an audio recording of the MRI scanner was played to simulate meditating in an MRI environment. Participants were instructed not to practice meditation outside of study training.

Post-intervention neuroimaging session

Study 1 and 2: After four, 20-minute intervention sessions, participants underwent fMRI scanning as the noxious 49°C series was delivered to the back of their right calf (See Figure 1 for study design). In the first two 49°C heat series, participants were instructed to “rest” (condition) with their eyes closed and pain ratings were collected after each series. After the first two heat series, anatomical acquisition was performed (MPRAGE) and participants in the mindfulness meditation and sham-mindfulness meditation groups (condition) were instructed to “begin practicing meditation” and to “meditate until the end of the study”. Two 49°C heat series were administered during meditation. The book-listening control group was instructed to “continue resting” with eyes closed during the two subsequent 49°C heat series and pain ratings were then collected. Study 2: During anatomical acquisition, the placebo cream was applied and removed after 10 minutes, [see (13) for more details] and two more noxious 49°C heat series were delivered to the area where the cream was administered and pain ratings were collected after each series.

Behavioral Pain Analyses

Pain intensity and unpleasantness ratings were entered into two independent generalized linear mixed models (LMM; “lme4” R package v1.1_35.3), fit using maximum-likelihood estimation. Fixed effects included the categorical predictors: Fixed effects included the categorical predictors of group, condition and sex, and all two- and three-way interactions. Models controlled for pre-intervention pain scores as a covariate. Random effects accounted for repeated measures and the nested structure of combining two studies, so random intercepts were included for subject and study-type. ANOVA tables were estimated with F-tests (and Satterthwaite approximated degrees of freedom) across all main effects and interactions. Significant F-tests ($p < 0.05$) were interpreted with post-hoc comparisons (“emmeans” R package v1.10.3) between marginal estimates for each factor level, including individual group X condition interactions. The Benjamini-Hochberg procedure was applied to control the false discovery rate (FDR) of multiple post-hoc comparisons.

Pain Signature Response Calculation

The Cognitive and Affective Neuroscience Laboratory

(https://github.com/canlab/Neuroimaging_Pattern_Masks) supplied pretrained and validated pain-predictive signature maps (23-25). All signatures were derived from pain-evoked fMRI activation maps and consisted of voxel-level regression weights optimized to predict pain ratings. These weighted spatial maps were obtained using machine learning methods and can be interpreted as the magnitude and direction of each voxel’s contribution to the prediction of pain. For each respective study, BOLD activation and CBF maps (input maps) were resampled to match the voxel dimensions of each pain signature. Then, the input and signature maps were flattened into vectors to compute the dot product between each input-signature pair (Nilearn Python package v0.9.1). This process yielded scalar values representing pattern similarity between each input-signature pair, which we refer to as “pain signature response” in the present paper. To equalize scale differences in pain signature responses between the two studies, thereby enabling comparisons of group and condition factors, the pain signature responses were standardized across all group and conditions within each respective study.

Pain Signature Response Analyses

Standardized NPS, NAPS, and SIIPS-1 responses were entered into three independent generalized linear mixed models (LMM; “lme4” R package v1.1_35.3), fit using maximum-likelihood estimation. Fixed effects included the categorical predictors: group X condition X sex and their interactions. Random effects accounted for repeated measures and the nested structure of combining two studies, so random intercepts were included for subject and study-type. ANOVA tables were estimated with F-tests (and Satterthwaite approximated degrees of freedom) across all main effects and interactions. Significant F-tests ($p < 0.05$) were interpreted with post-hoc comparisons (“emmeans” R package v1.10.3) between marginal estimates for each factor level, including individual group X condition interactions. The Benjamini-Hochberg procedure was applied to control the false discovery rate (FDR) of multiple post-hoc and correlation comparisons. Additionally, pairwise correlations between manipulation-driven changes in pain and pain signature responses were estimated (see corresponding methods and results in Supplementary appendix).

RESULTS

Mindfulness meditation is more effective than placebo-cream and sham-mindfulness meditation at reducing pain.

Pain Intensity: The group X condition interaction, $F(6, 641.15) = 12.58, p < .001$, was associated with the significant mindfulness meditation induced pain intensity reductions when compared to the placebo-cream ($p = .007$), sham-mindfulness meditation ($p = .002$) and controls ($p < .001$) (Figure S1 left; see Table S1 for mean ratings and Table S2 for complete model). The placebo-cream ($p < .001$) and sham-mindfulness meditation ($p < .001$) was also associated with pain intensity reductions when compared to the controls. There were no significant differences between the sham-mindfulness meditation and placebo-cream groups on pain intensity reductions ($p = .58$; Figure S1).

Pain Unpleasantness: The group X condition interaction, $F(6, 640.96) = 14.97, p < .001$, was associated with the significant mindfulness meditation-induced pain unpleasantness reductions when compared to the placebo-cream ($p = .003$), sham-mindfulness meditation ($p = .05$) and controls ($p < .001$) (Figure S1 right; see Table S1 for mean ratings and Table S3 for full model). Placebo-cream ($p < .001$) and sham-mindfulness meditation ($p = .003$) also significantly lowered pain unpleasantness ratings when compared to the controls. There were no significant differences between the sham-mindfulness meditation and placebo-cream groups on pain unpleasantness ratings ($p = .34$).

Mindfulness meditation reduces nociceptive-specific pain signatures when compared to placebo-cream, sham-mindfulness meditation, and controls.

The significant group X condition interaction on NPS, $F(3, 338.30) = 11.65, p < .001$ was associated with the significant reductions in NPS responses during mindfulness meditation when compared to placebo-cream ($p = .004$), sham-mindfulness meditation ($p = .003$), and book-listening control ($p < .001$) groups (Figure 2; Table S4). Mindfulness meditation was associated with significant NPS reductions from rest to meditation ($p < .001$; $CI95 = -.87, -.49$). The placebo-cream, sham-mindfulness meditation and controls did not significantly modify the NPS responses ($ps = .97$). There were no other significant between group differences (see Table S4 for complete NPS LMM results).

Mindfulness meditation reduces negative affective pain signatures when compared to placebo and controls.

The significant group X condition interaction on negative affective pain signatures, $F(3, 338.55) = 4.69, p = .003$ was associated with significantly greater NAPS reductions exhibited during mindfulness meditation when compared to placebo-cream ($p = .04$) and control ($p = .04$) conditions (Figure 3; Table S5) but not sham-mindfulness meditation ($p = .30$). Mindfulness meditation significantly reduced NAPS responses from rest to meditation ($p < .001$; $CI95 = -.67, -.29$). However, neither placebo-cream ($p = .81$), sham-mindfulness meditation ($p = .23$) nor the controls ($p = .73$) significantly modulated NAPS. There were no other significant between group differences (see Table S5 for complete NAPS LMM results).

Placebo-cream reduces the stimulus intensity independent pain signature.

The interaction on the placebo-related pain signatures (SIIPS-1), $F(3, 338.21) = 5.13, p = .002$ was associated with significant SIIPS-1 reductions in the placebo-cream group when compared to rest ($p < .05$; $CI95 = -.56, -.11$; Figure 4; Table S6). Sham-mindfulness meditation significantly increased SIIPS-1 responses ($p < .05$; $CI95 = -.56, -.13$). Neither mindfulness meditation ($p = .81$) nor controls ($p = .36$) significantly modulated SIIPS-1. Placebo-cream produced significant reductions in SIIPS-1 responses when compared to sham-mindfulness meditation ($p = .01$; $CI95 = -1.02, -.24$). There were no other significant between group differences (see Table S6 for complete SIIPS-1 LMM results).

DISCUSSION

Psychological and behavioral strategies are increasingly used for treating pain (29) because they offer a safer and more reliable alternative to opioids (30). Justification for placebo-controls is based on the underlying assumption that placebo engages a subset of mechanisms overlapping *active* and non-inert treatments (31). Although it has long been assumed that mind-body approaches like meditation merely reflect placebo, here we observe a double dissociation in behavioral and neural mechanisms supporting mindfulness meditation and placebo. Mindfulness meditation produced significantly greater reductions in nociceptive-specific and negative-affect pain signature responses. In contrast, placebo-cream significantly reduced pain and only attenuated the SIIPS-1 signature. This double dissociation indicates that brain systems underlying mindfulness versus placebo analgesia are distinct and demonstrates granularity in pain-related modulation between the two manipulations. Prior work has demonstrated that mindfulness meditation largely impacts pain unpleasantness and not pain intensity (32). In the present study, mindfulness meditation was associated with a modulation of pain intensity *and* pain unpleasantness ratings (Figure S1). Thus, mindfulness meditation can comprehensively impact sensory and more affectively aligned dimensions of the pain experience. In contrast, placebo-cream and sham-mindfulness meditation only significantly reduced pain unpleasantness ratings. The control group exhibited a significant increase in pain intensity and unpleasantness from the first half to the second half of the scan (i.e., manipulation) (Figure S1), likely indicating higher pain sensitization from repeated stimulation to the back of the calf while lying in the scanner.

Importantly, we provide novel evidence, across two studies with different fMRI techniques, that mindfulness meditation significantly reduces pain signatures supporting nociceptive-driven and negative affective dimensions of pain. This is also the first study to validate the utility of pain signatures using perfusion-based fMRI. Correspondence between pain signature responses and subjective pain reports showed higher NPS responses were associated with higher pain intensity and unpleasantness ratings (Table S7). In contrast, the NAPS was only positively associated with pain unpleasantness ratings. Placebo-cream significantly reduced the SIIPS-1, but no other pain signatures, providing credence that the SIIPS-1 represents neural processes supporting placebo. Sham-mindfulness meditation provided an operational comparator to mindfulness meditation to better isolate the active mechanisms specific to mindfulness (non-reactive awareness) while controlling for the non-specific factors engaged by meditation (breathing; posture). Surprisingly, sham-mindfulness meditation significantly increased SIIPS-1 responses. In prior work conducted in healthy individuals, sham-mindfulness meditation-induced analgesia was found to be opioidergically-mediated (33) and associated with higher PAG activation (13). Thus, higher SIIPS-1 during sham-mindfulness meditation may reflect engagement of placebo-related processing.

The NPS is representative of sensory (thalamus, posterior insula) processing and the NAPS is sensitive to responses supporting shared aversiveness between nociceptive and negative affective processing (precuneus; amygdala; anterior mid-cingulate; brainstem), brain mechanisms that also mediate mindfulness meditation based pain and anxiety relief (11-13). Thus, our present findings corroborate our working theoretical hypotheses (34) that mindfulness meditation-induced pain relief engages multiple pain modulatory mechanisms supporting volitionally mediated disengagement from self-referential appraisals of nociceptive input that do not overlap with those engaged by placebo. Taken together, we provide complementary evidence to a growing body of work demonstrating that placebo-based analgesia does not simply reflect a *shared* subset of mechanisms with active treatments like meditation, and that different processes impact unique neural systems. This distinction confirms the existence of multiple supraspinal mechanisms for pain modulation and is analogous to recent work showing that a placebo cream reduces SIIPS-1 but not NPS (35). This is the first known study to demonstrate modulation of the NAPS by any therapeutic approach. In the present study, neither placebo-cream nor sham-mindfulness meditation significantly modulated the NAPS. Thus, the NAPS may be a unique and reliable marker of non-placebo based pain modulation that may be

particularly useful for researching chronic pain populations that have exacerbated pain unpleasantness as compared to pain intensity (36). A recent study found that, during rest (i.e., non-meditation) and noxious heat, an 8-week Mindfulness-Based Stress Reduction program did not significantly reduce pain in healthy individuals, NPS or SIIPS-1 responses when compared to waitlisted controls during rest (i.e., non-meditation) (37). However, the effects of active meditation on NPS and SIIPS-1 responses during painful heat were not tested and the NAPS was not employed. Although pain ratings were not collected, another study (38), in 17 healthy individuals, found that the instruction “to attend to and accept their experience as it is” was associated with lower negative affect behavioral reports and NPS responses to thermal stimulation when compared to instructions to “react naturally, whatever your response might be.” The present results similarly are not explicitly generalizable to patients living with chronic pain because they were performed in healthy individuals. We also cannot explicitly deduce whether changes in pain signatures were related to mental training/placebo because fMRI was not acquired in the pre-intervention testing sessions. Further, larger sample sizes could enhance the interpretability of the present results.

As previously described in contemporary and traditional texts, mindfulness meditation may reduce pain by decoupling egocentric appraisals of salient, nociceptive processes. Mindfulness meditation based analgesia has been found to decouple midline cortical structures (mPFC) supporting self-referential processing with the thalamus, a low-level nociceptive relay mechanism (11). The present findings support these prior findings in demonstrating that mindfulness meditation significantly reduces the NAPS, which includes pain-related processing in the thalamus and mPFC. This is important because chronic pain development and exacerbations are largely driven by affective dysregulation (39). Thus, the present findings provide insight into demonstrating that mindfulness meditation modulates nociceptive specific and negative affective mechanisms that may be reflective of a mindfulness meditation-based filtering mechanism supporting shifts in attentional engagement and attenuation of affective appraisals of noxious stimuli (40). In summary, we show that mindfulness meditation is more effective at reducing pain and mechanistically distinct from placebo. This distinction confirms the existence of multiple, cognitively driven supraspinal mechanisms for pain modulation via mindfulness meditation and may foster its greater acceptance as an adjunct pain therapy.

FIGURE LEGENDS

Figure 1: Experimental Design and Multivariate Analyses.

a. Overview of the experimental design. Across two clinical trials, and after baseline pain testing, participants were randomized to a four session, 20-minute mindfulness meditation, sham-mindfulness meditation, active book-listening or placebo-cream control regimen. Following the interventions, participants rested with their eyes closed during fMRI acquisition and noxious thermal stimulation to their right calf. For the manipulation in the second half of the neuroimaging session, the individuals in the mindfulness and sham-mindfulness meditation were instructed to meditate, the placebo group was applied (and removed) placebo “lidocaine” cream (petroleum jelly), and the book-listening group continued to rest. Visual analog pain intensity and unpleasantness ratings (0 = “no pain” to 10 = “worst pain imaginable”) were collected after each run. **b.** Previously published and validated multivariate fMRI pain-predictive signatures and corresponding pattern weights. Insets indicate individual voxel weights $w_1, w_2, \dots, w_n$ represented as vectors. **c.** Pain signature responses were computed for each of the pain-predictive signatures for all imaging series. The pain predictive signatures comprised of voxel weights that specified how to integrate brain data to produce a pain signature response (i.e., predicted outcome). Calculating the dot product between the predictive map and test images—weighted sum of activity across the test images, with the predictive map specifying the weights—generates a pain signature response. Study 1 test images were thermal-evoked BOLD beta estimates (X_{BOLD}) corresponding to the contrast [49°C - 35°C]. Study 2 test images were thermal-evoked cerebral blood flow responses (X_{CBF} in ml/100g/min) averaged across the stimulus trials.

Figure 2: Mindfulness meditation produced greater reductions in nociceptive-specific pain signatures when compared to placebo and the controls.

***Mindfulness meditation elicited significant reductions in the nociceptive specific pain signature (NPS) when compared to the placebo-cream ($p = .004$), sham-mindfulness meditation ($p = .003$) and book-listening control ($p < .001$) conditions.

Figure 3: Mindfulness meditation produced greater reductions in negative affective pain signatures when compared to placebo, sham-mindfulness meditation and the controls.

***Mindfulness meditation was associated with significantly greater reductions in the NAPS when compared placebo cream ($p = .04$) and control ($p = .04$) conditions, but not sham-mindfulness meditation ($p = .30$). Mindfulness meditation also significantly reduced NAPS responses from rest to meditation ($p < .001$). Neither placebo-cream ($p = .81$), sham-mindfulness meditation ($p = .23$) nor the controls ($p = .73$) significantly modulated the NAPS.

Figure 4: Placebo-cream was associated with lower stimulus independent pain signature responses.

*Placebo-cream ($p < .05$) significantly lowered SIIPS-1 responses when compared to rest. In contrast, **sham-mindfulness meditation significantly increased SIIPS-1 responses ($p < .05$). Neither mindfulness meditation ($p = .81$) nor controls ($p = .36$) significantly modulated SIIPS-1. Placebo-cream significantly lowered SIIPS-1 responses when compared to sham-mindfulness meditation ($p = .01$).

FINANCIAL DISCLOSURES

The authors report no biomedical financial interests or potential conflicts of interest.

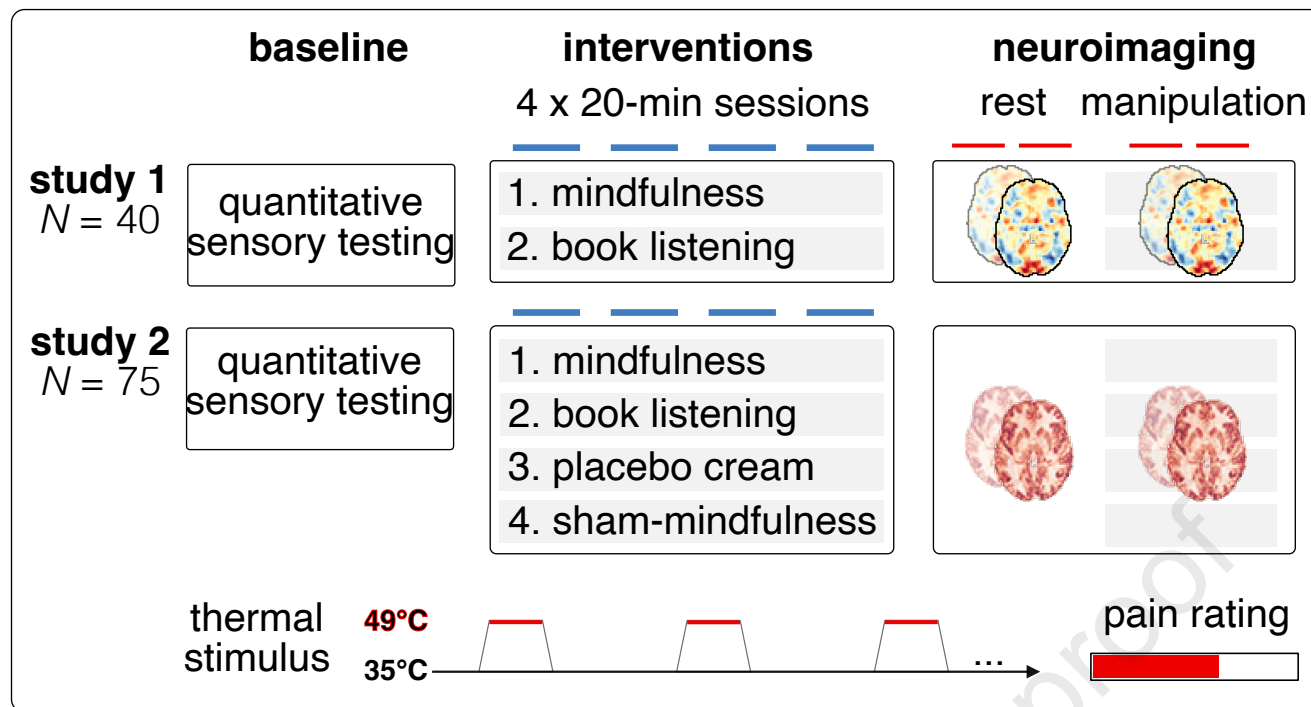
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REFERENCES

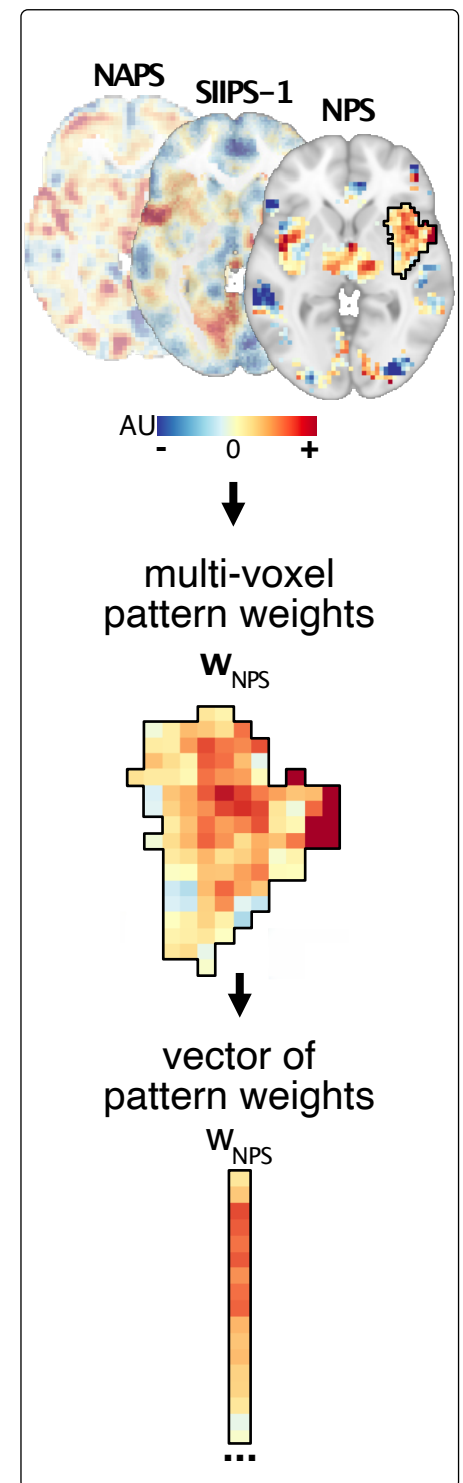
1. Wager TD, Nitschke JB (2005): Placebo effects in the brain: linking mental and physiological processes. *Brain Behav Immun*. 19:281-282.
2. Benedetti F, Mayberg HS, Wager TD, Stohler CS, Zubieta JK (2005): Neurobiological mechanisms of the placebo effect. *J Neurosci*. 25:10390-10402.
3. Colloca L, Barsky AJ (2020): Placebo and Nocebo Effects. *N Engl J Med*. 382:554-561.
4. Garland EL, Hanley AW, Nakamura Y, Barrett JW, Baker AK, Reese SE, et al. (2022): Mindfulness-Oriented Recovery Enhancement vs Supportive Group Therapy for Co-occurring Opioid Misuse and Chronic Pain in Primary Care: A Randomized Clinical Trial. *JAMA Intern Med*.
5. Wells RE, O'Connell N, Pierce CR, Estave P, Penzien DB, Loder E, et al. (2021): Effectiveness of Mindfulness Meditation vs Headache Education for Adults With Migraine: A Randomized Clinical Trial. *JAMA Intern Med*. 181:317-328.
6. Cherkin DC, Sherman KJ, Balderson BH, Cook AJ, Anderson ML, Hawkes RJ, et al. (2016): Effect of Mindfulness-Based Stress Reduction vs Cognitive Behavioral Therapy or Usual Care on Back Pain and Functional Limitations in Adults With Chronic Low Back Pain: A Randomized Clinical Trial. *JAMA*. 315:1240-1249.
7. Vajirā, Maha Bodhi society Calcutta. (1941): *The Sutta-nipāta, one of the oldest canonical books of the Buddhists ... (English translation with Pāli text)*. Sarnath, U.P.; Maha Bodhi society.
8. Lutz A, McFarlin DR, Perlman DM, Salomons TV, Davidson RJ (2013): Altered anterior insula activation during anticipation and experience of painful stimuli in expert meditators. *Neuroimage*. 64:538-546.
9. Grant JA, Courtemanche J, Rainville P (2011): A non-elaborative mental stance and decoupling of executive and pain-related cortices predicts low pain sensitivity in Zen meditators. *Pain*. 152:150-156.
10. Brown CA, Jones AKP (2010): Meditation experience predicts less negative appraisal of pain: electrophysiological evidence for the involvement of anticipatory neural responses. *Pain*. 150:428-438.
11. Riegner G, Posey G, Oliva V, Jung Y, Mobley W, Zeidan F (2023): Disentangling self from pain: mindfulness meditation-induced pain relief is driven by thalamic-default mode network decoupling. *Pain*. 164:280-291.
12. Zeidan F, Martucci KT, Kraft RA, Gordon NS, McHaffie JG, Coghill RC (2011): Brain mechanisms supporting the modulation of pain by mindfulness meditation. *J Neurosci*. 31:5540-5548.
13. Zeidan F, Emerson NM, Farris SR, Ray JN, Jung Y, McHaffie JG, et al. (2015): Mindfulness Meditation-Based Pain Relief Employs Different Neural Mechanisms Than Placebo and Sham Mindfulness Meditation-Induced Analgesia. *J Neurosci*. 35:15307-15325.
14. Finan PH, Hunt C, Keaser ML, Smith K, Lerman S, Bingham CO, et al. (2024): Effects of Savoring Meditation on Positive Emotions and Pain-Related Brain Function: A Mechanistic Randomized Controlled Trial in People With Rheumatoid Arthritis. *J Pain*.
15. Roy M, Shohamy D, Wager TD (2012): Ventromedial prefrontal-subcortical systems and the generation of affective meaning. *Trends in Cognitive Sciences*. 16:147-156.
16. Khatib L, Dean JG, Oliva V, Riegner G, Gonzalez NE, Birenbaum J, et al. (2023): The role of endogenous opioids in mindfulness and sham mindfulness-meditation for the direct alleviation of evoked chronic low back pain: a randomized clinical trial. *Neuropsychopharmacology*.
17. Wells RE, Collier J, Posey G, Morgan A, Auman T, Strittmatter B, et al. (2020): Attention to breath sensations does not engage endogenous opioids to reduce pain. *Pain*. 161:1884-1893.
18. May LM, Kosek P, Zeidan F, Berkman ET (2018): Enhancement of Meditation Analgesia by Opioid Antagonist in Experienced Meditators. *Psychosom Med*.
19. Levine JD, Gordon NC, Fields HL (1978): The mechanism of placebo analgesia. *Lancet*. 2:654-657.
20. Zunhammer M, Spisák T, Wager TD, Bingel U, Consortium PI (2021): Meta-analysis of neural systems underlying placebo analgesia from individual participant fMRI data. *Nat Commun*. 12:1391.

21. Hoepfli ME, Nahman-Averbuch H, Hinkie WA, Leon E, Peugh J, Lopez-Sola M, et al. (2022): Dissociation between individual differences in self-reported pain intensity and underlying fMRI brain activation. *Nat Commun.* 13:3569.
22. Reddan MC, Wager TD (2018): Modeling Pain Using fMRI: From Regions to Biomarkers. *Neurosci Bull.* 34:208-215.
23. Wager TD, Atlas LY, Lindquist MA, Roy M, Woo CW, Kross E (2013): An fMRI-based neurologic signature of physical pain. *N Engl J Med.* 368:1388-1397.
24. Woo CW, Schmidt L, Krishnan A, Jepma M, Roy M, Lindquist MA, et al. (2017): Quantifying cerebral contributions to pain beyond nociception. *Nat Commun.* 8:14211.
25. Čeko M, Kragel PA, Woo CW, López-Solà M, Wager TD (2022): Common and stimulus-type-specific brain representations of negative affect. *Nat Neurosci.* 25:760-770.
26. Zunhammer M, Bingel U, Wager TD, Consortium PI (2018): Placebo Effects on the Neurologic Pain Signature: A Meta-analysis of Individual Participant Functional Magnetic Resonance Imaging Data. *JAMA Neurol.* 75:1321-1330.
27. Botvinik-Nezer R, Petre B, Ceko M, Lindquist MA, Friedman NP, Wager TD (2024): Placebo treatment affects brain systems related to affective and cognitive processes, but not nociceptive pain. *Nat Commun.* 15:6017.
28. White G (1789): *The Natural History of Selborne*. Benjamin White.
29. Pritchard KT, Baillargeon J, Lee WC, Raji MA, Kuo YF (2022): Trends in the Use of Opioids vs Nonpharmacologic Treatments in Adults With Pain, 2011-2019. *JAMA Netw Open.* 5:e2240612.
30. Krebs EE, Gravely A, Nugent S, Jensen AC, DeRonne B, Goldsmith ES, et al. (2018): Effect of Opioid vs Nonopioid Medications on Pain-Related Function in Patients With Chronic Back Pain or Hip or Knee Osteoarthritis Pain: The SPACE Randomized Clinical Trial. *JAMA.* 319:872-882.
31. Burke MJ, Romanella SM, Mencarelli L, Greben R, Fox MD, Kaptchuk TJ, et al. (2022): Placebo effects and neuromodulation for depression: a meta-analysis and evaluation of shared mechanisms. *Mol Psychiatry.* 27:1658-1666.
32. Perlman DM, Salomons TV, Davidson RJ, Lutz A (2010): Differential effects on pain intensity and unpleasantness of two meditation practices. *Emotion.* 10:65-71.
33. Wells RE, Collier J, Posey G, Morgan F, Auman T, Strittmatter B, et al. (2020): Attention to breath sensations does not engage endogenous opioids to reduce pain. *Pain.*
34. Grant JA, Zeidan F (2019): Employing pain and mindfulness to understand consciousness: a symbiotic relationship. *Curr Opin Psychol.* 28:192-197.
35. Botvinik-Nezer R, Petre B, Ceko M, Lindquist MA, Friedman NP, Wager TD (2023): Placebo treatment affects brain systems related to affective and cognitive processes, but not nociceptive pain. *bioRxiv.*
36. Price DD, Harkins SW, Baker C (1987): Sensory-affective relationships among different types of clinical and experimental pain. *Pain.* 28:297-307.
37. Wielgosz J, Kral TRA, Perlman DM, Mumford JA, Wager TD, Lutz A, et al. (2022): Neural Signatures of Pain Modulation in Short-Term and Long-Term Mindfulness Training: A Randomized Active-Control Trial. *Am J Psychiatry.* 179:758-767.
38. Kober H, Buhle J, Weber J, Ochsner KN, Wager TD (2019): Let it be: mindful acceptance down-regulates pain and negative emotion. *Soc Cogn Affect Neurosci.* 14:1147-1158.
39. Hashmi JA, Baliki MN, Huang L, Baria AT, Torbey S, Hermann KM, et al. (2013): Shape shifting pain: chronification of back pain shifts brain representation from nociceptive to emotional circuits. *Brain.* 136:2751-2768.
40. Garland EL, Roberts RL, Hanley AW, Zeidan F, Keefe FJ (2023): The Mindful Reappraisal of Pain Scale (MRPS): Validation of a New Measure of Psychological Mechanisms of Mindfulness-Based Analgesia. *Mindfulness (N Y).* 14:192-204.

A. Experimental Design



B. Pain Signatures



C. Pain Signature Responses

